

Combining Data Centers with Electric Vehicle Battery Swapping Stations for Grid Regulation

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Abstract—The combination of data center servers and electric vehicles (EVs) for grid regulation has attracted more attention in recent years. However, state of the art solutions have focused on utilizing charging stations but ignored the emerging battery swapping stations. In this paper, we argue that EV battery swapping stations may play an important role in smart grid regulation as well, especially when combining with clusters of data center servers. We propose a novel integrated system that can optimize the regulation capacity while maintaining the quality of the service provided by battery swapping station. Our strategy incorporates novel algorithms that leverage historical information in decision making process. Simulation results demonstrate that our solution reduces the imbalance of grid by 80 ~ 85% compared with the baseline approach, and by 50 ~ 70% compared with the other alternative approach.

Keywords—grid regulation, battery swapping station, data center servers

I. INTRODUCTION

As the concept of smart grid appears and importance of electricity resource grows, new methods to better utilize electricity produced by power plants are emerging. In recent years, adaptive control of data center servers [1] and electric vehicles for grid regulation has attracted more attentions. Data centers account for 1.5% of total energy demand in the United States [2]. EVs are anticipated to alleviate the environment crisis. The demand and supply of electric power are usually imbalanced and unsteady, and a traditional grid structure cannot leverage the redundant power or complement the insufficient part of power. Besides, renewable power such as wind and solar energy are encouraged to be used globally, while increasing the fluctuation of the grid. Thus, the regulating ability of power consumption by data centers and the power storage ability of electric vehicles will help stabilize and benefit the grid a lot. Data centers can change the power usage profile by increasing or decreasing the frequency of server CPU to an extent, while batteries in EV refueling stations can store or discharge electricity when plugged in grid.

There have been previous research on regulation assets participating in grid regulation. [3] showed how data centers stabilize grid. [4] presented a conceptual framework to successfully integrate electric vehicles into electric power systems. [5] has proved vehicle-to-grid power (V2G) is efficient

in stabilizing the grid and supporting large-scale renewable energy. [6] used electric vehicle charging as a way to help offset the imbalance associated with penetration of renewable generation. [7] scheduled a smart charge of plug-in hybrid electric vehicle (PHEV) batteries to minimize the fluctuations of wind energy. [8], [9] explored the regulation ability of dynamic voltage and frequency scaling (DVFS) of servers in data centers. The results of [10] indicated that the regulation signal of DVFS can be tracked very accurately and a single data center can achieve about a half-million-dollar revenue by participating in the regulation market without affecting its service quality. [11] proposed an algorithm about how EVs in battery charging station maximize their profit. [12] combined data center and EVs in battery charging station to maximize the profit of data centers, showed how regulation market operates in practice. There are also some papers considering battery swapping station participating in grid regulation. [13] presented a solution concentrating on unit commitment. [14] showed a strategy through a method called adaptive mutation particle swarm optimization.

However, state of the art solutions have focused on the EV charging stations but ignored the emerging battery swapping stations combining with data center. We argue that since the charging station only has batteries when EVs are charging in that station, and the drivers of the EVs usually expect to get the EVs fully charged as soon as possible, the capacity of regulation produced by charging station is strictly limited. On the contrary, battery swapping stations always contain a certain number of batteries, the capacity of regulation is proportional to the total energy capacity of all batteries in the station. In this paper, we propose a novel integrated system that optimally utilizes the electricity regulation capacity of battery swapping station and data center to stabilize the grid. The challenge lies on maintaining a certain service quality of swapping station without a prior knowledge about coming EVs under the constraint of limited batteries in the station. Our strategy incorporates novel algorithms that leverage historical information in decision making process. Simulation results demonstrate that our solution reduces the imbalance of grid by 80 ~ 85% compared with the baseline approach, and by

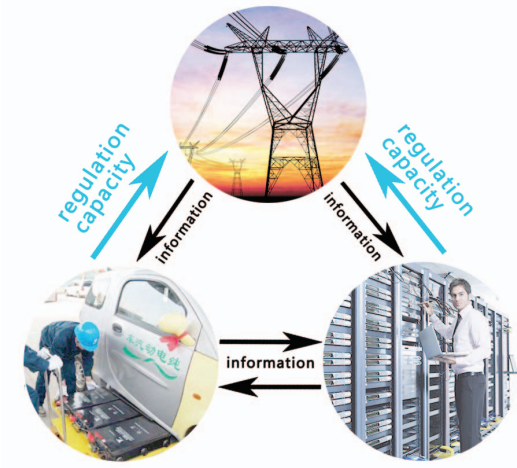


Fig. 1. System overview.

50 ~ 70% compared with the other alternative approach.

The contribution of this paper is mainly three-fold: 1) to the best of our knowledge, this is the first paper that combines data centers with electric vehicle battery swapping stations for grid regulation; 2) we propose new methods to optimize the regulation capacity while maintaining the service quality of battery swapping station; and 3) simulation results show that our approach outperforms alternative algorithms in the perspective of stabilizing the electric grid.

The paper is organized as follows. Section II gives our model and assumption. Our methods and algorithms are illustrated in Section III. Section IV evaluates the results of simulation and section V concludes our paper.

II. SYSTEM MODEL

In this section, we describe our mathematical framework of our problem, including the system model and problem statement.

A. Model and Assumptions

In the grid regulation market, grid regulation is composed of two processes [15]. As shown in Fig.1, at the beginning of each operation hour, the regulation assets send their capacity ranges in this cycle to the grid (controller), then in the operating hour, the regulation assets deal with the electric energy that the grid claimed. The given energy is restricted in the range given in the first process. Our model is designed to calculate an energy regulation interval which data centers and swapping stations can reach to balance the generation and consumption of grid as much as possible.

We start with describing the battery swapping stations. A battery swapping station always has a certain number of batteries, namely N batteries can be used in regulation. According to the industry data, the battery can usually be charged to almost full in about two hours. Thus to display the complexity of charging in multiple hours, we set 2 hours as the default net time the battery will be charged, while the fully discharging time is B

hours. Besides, we denote the electric energy which batteries in battery swapping station have equals to the energy of $c(i)$ EVs' batteries before the i th operating hour ($i = \{1, 2, \dots, 24\}$). We denote $c(1)$ as c , which is the primitive status in the beginning of the 24 hour cycle.

According to the mechanism of grid regulation, the process is in one hour cycle. According to our research on the charging pattern in Guangzhou, the need of battery charging in one day share the similar pattern to that in another day, so as the pattern of grid. Thus it is natural to suppose they are in the normal distribution. Note the electric energy expected to be hold by battery swapping station or other electric terminals of each hour in a day equals to the electric energy of $m(i)$ EVs' batteries, and its distribution satisfies $N(u(i), \sigma_u(i))$. The number of batteries exchanged in each hour of the day is $n(i)$, and its distribution satisfies $N(v(i), \sigma_v(i))$. The distribution pattern can also be deduced by more advanced statistical tools such as time series analysis, multivariate analysis or regression analysis.

Swapping station should guarantee a promised service quality while provide regulation capacity, in our model, it should allow a proportion α of EVs getting batteries exchanged without waiting.

In the best case, the regulation capacity should cover the redundant or the shortage of electricity, thus we use the concept Excessive Power Measurement (EPM) denoted by L to indicate the insufficient part of the regulation capacity. Since dealing with the excessive power will become harder for other electrical terminals when the redundant energy is larger, we use $\int (g(t) - h(t))^2$ to denote EPM, $g(t)$ refers to the electric energy expected to be hold by battery swapping station, and $h(t)$ refers to the electric energy the battery swapping station hold in practice. The electricity held by swapping station in each hour is equal to the energy of $k(i)$ batteries. So the EPM in one hour considering swapping station is

$$l(i) = (m(i) - k(i))^2$$

Then we illustrate the role of data center in our method. Data centers send their baseload and regulation capacity to the grid in the beginning of operating hour. Baseload is nearly the average power consumption in each hour, and regulation capacity is the range of consumption change. Data centers participate in grid regulation mainly through adjusting the frequency of servers, so we can assume the regulation capacity of data center are independent between hours. We denote the regulation capacity of each hour by $h(i)$. In this case, we get

$$l(i) = (\max(|k(i) - m(i)|, h(i)) - h(i))^2$$

which is the EPM in one hour contains both swapping station and data center. The aim of our paper is to find the best parameters $k(i)$ to make EPM as small as possible.

B. Problem Statement

In our model, we aim at minimizing the EPM, thus the optimization function is:

$$\min \sum_{i=1}^{24} l(i)$$

The objective is to reduce the load burden of other electric terminals which need to tackle the excessive power, smaller L means larger regulation capacity of our method. The constraints and details will be discussed in next section.

III. ADAPTIVE SCHEDULING ALGORITHM

In this section, we compare several strategies that can be applied to minimize the imbalance of grid.

A. Overview

We introduce several strategies about power regulation. Baseline and bestline are reference of our strategies to test their effect. Baseline is the result when battery swapping station don't participate in grid regulation. In other words, the station keeps charging batteries which are not full all the time. Bestline assumes to know the actual data rather than distributions, so the optimal solution can be applied, which cannot be achieved in practice. The problem here is choosing $k(i)$ to minimize EPM while having known all related data $m(i)$, $n(i)$ before the 24 hour cycle starts. Present strategy gives the interval of each hour only considering the physical restriction. Expectation strategy assume the uncertain data equals its expectation, and gives its bounds of this operating hour which maximize the regulation capacity of the whole day. Mild strategy and deviation strategy are modification of above strategies, which is discussed in detail when they are applied.

The procedure of our method is as following: first we use history information to calculate the data distribution, second we choose one strategy to decide the constraints, then the optimal charging strategy $k(i)$ is calculated under the constraints deduced in second step.

The followings are common restrictions:

- 1) Battery finishes its charge in 2 hours according to our assumption, so we have to consider charging conditions in the next two hour while considering the condition in this hour. The condition should satisfy: $n(i) + 3n(i+1)/4 + n(i+2)/4 \leq c(i) + k(i)$.
- 2) The electric energy the swapping station have before each operation hour can not be less than zero or larger than full, so we get $0 \leq c(i) \leq N$.
- 3) Energy conservation: $c(i+1) = c(i) + k(i) - n(i)$.

In order to clarify our algorithm, we introduce this without considering data center servers at first. Then we modify the algorithm with data center servers introduced (DVFS).

B. Present Strategy

This strategy ignores the information about from history data. The station sends the regulation capacity in each hour as large as possible to the grid. The following are the discussion about the restrictions:

- 1) $n(i)$ is not a prior knowledge in practice when the operating hour begins. To guarantee the quality of battery swapping service, we have: $d(i) + 3d(i+1)/4 + d(i+2)/4 \leq c(i) + k(i)$. Among them, $a = \text{norm}(\alpha)$, $b(i) = \sigma_v(i, i)$, $d(i) = v(i) + b(i) * a$.
- 2) m is also not a prior knowledge in practice when the operating hour begins. To guarantee that battery upper bound can be reached in practice, we can get: $c(i) + k(i) - v(i) + b(i) * a \leq N$.
- 3) It takes 2 hours to charge batteries to be full, so we have $k(i) \leq N/2$. It takes B hours to discharge full batteries to be empty, we can get $-k(i) \leq N/B$. Thus $-N/B \leq k(i) \leq N/2$.

Finally, we get $k(i)$ through the following formula:

We send the interval of $[max(-N/B, d(i) + 3d(i+1)/4 + d(i+2)/4 - c(i)), min(N/2, N + v(i) - b(i) * a - c(i))]$ to the grid in the beginning of the operating hour, $k(i)$ get the value which has the least distance to $m(i)$ in the interval.

There is no difference in deciding $k(i)$ when data center servers are introduced in present strategy.

1) *Mild Strategy*: In the present strategy, we give the largest interval as much as possible. However, we may get a better result if we reserve some regulation capacity to the next hour. So Mild Strategy is introduced based on the previous Present Strategy. We get the interval $[PreMin, PreMax]$, and then send the interval $[\beta PreMin + (1 - \beta) PreMax, \beta PreMax + (1 - \beta) PreMin]$ to the grid in the beginning of the operating hour, $k(i)$ get the value which has the least distance to $m(i)$ in the interval.

There is no difference in deciding $k(i)$ when data center servers are introduced in mild strategy.

C. Expectation Strategy

Assuming the related unknown data equal to the expected value, we can get the solution of the operating hour in the model of the optimization problem. We can get two bounds through choosing two different objective functions:

- 1) Choose $k1(i)$ to minimize the objective function including this hour and later in the day, the objective function is $\sum_{j=i}^{24} l(j)$
- 2) Choose $k2(i)$ to minimize the objective function of later hours in the day, the objective function is $\sum_{j=i+1}^{24} l(j)$

Finally, we get $k(i)$ through the following formula, note that $i < j \leq 24$ and k_* means that $k1$ and $k2$ share the same form

of constraints:

$$\begin{aligned}
& \underset{k_1}{\text{minimize}} && \sum_{j=i}^{24} (u(j) - k_1(j))^2 \\
& \text{and} \\
& \underset{k_2}{\text{minimize}} && \sum_{j=i+1}^{24} (u(j) - k_2(j))^2 \\
& \text{s.t.} \begin{cases} d(i) + 3d(i+1)/4 + d(i+2)/4 \leq c(i) + k_*(i) \\ v(j) + 3v(j+1)/4 + v(j+2)/4 \leq c(j) + k_*(j) \\ c(i) + k_*(i) - v(i) + b(i) * a \leq N \\ c(j+1) = c(j) + k_*(j) - v(j) \\ -N/B \leq k_*(i) \leq N/2 \\ c(j) \leq N \end{cases}
\end{aligned}$$

We send the interval of $k_1(i)$ and $k_2(i)$ in the beginning of the operating hour to the grid, and $k(i)$ get the value which has the least distance to $m(i)$ in the interval.

When data center servers are introduced, we should choose the corresponding $l(i)$ at first, and choose $k(i)$ which is closest to $k_2(i)$ while the regulation capacity of data center servers is flexible.

1) *Deviation Strategy*: Considering the fluctuation of $m(i)$ based on Expectation Strategy, we may get a better result if we enlarge the interval on the basis of the interval deduced from Expectation Strategy. If $k(i)$ was in the previous interval, the result wouldn't change. However, if $k(i)$ wasn't contained in the previous interval, the regulation capacity at the present hour was enlarged, while the fluctuations of grid may make the loss of regulation capacity in the future hours be relatively smaller. So the objective function can be $\min(\sum_{j=i}^{24} (u_1(j) - k_1(j))^2, \sum_{j=i}^{24} (u_2(j) - k_2(j))^2, \sum_{j=i+1}^{24} (u(j) - k_3(j))^2)$.

Among them,

$$\begin{cases} u_1(i) = u(i) - d(i) * \sigma_u(i, i) \\ u_2(i) = u(i) + d(i) * \sigma_u(i, i) \\ u_1(j) = u_2(j) = u(j), j > i \end{cases} \quad (1)$$

The restriction condition is same as the expectation strategy. We get $k_1(i)$, $k_2(i)$, $k_3(i)$ through solving the optimization problem. And then, we send the interval of $[\min(k_1(i), k_2(i), k_3(i)), \max(k_1(i), k_2(i), k_3(i))]$ to the grid in the beginning of the operating hour, and $k(i)$ get the value which has the least distance to $m(i)$ in the interval.

When data center servers are introduced, we should choose the corresponding $l(i)$ in the algorithm at first, and choose $k(i)$ which is closest to $k_3(i)$ while the regulation capacity of data center servers is flexible.

IV. EVALUATION

In the section, we present the evaluation results. First, we examine the effect of different algorithm when parameters change to compare present strategy with expectation strategy in various conditions. Next, we compare our strategies with their modified version. Finally, we report the results of our strategy when data center servers in data center are introduced.

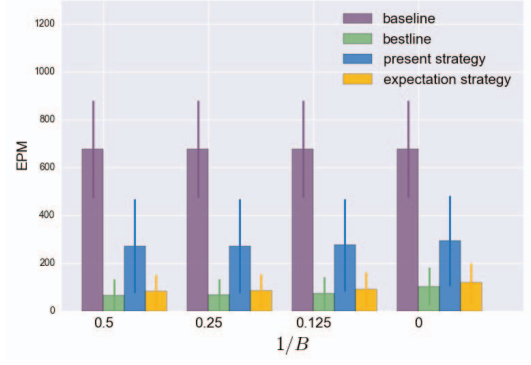


Fig. 2. EPM of varying $1/B$.

A. Simulation Setup

The electric energy that battery swapping station consume should equal to the the energy that is expected to be consumed to reach the balance of grid. For the convenience of computing, we note the value to be $5 * 24 = 120$. The conclusion can be applied to actual consequences no matter what value we choose in the simulation.

The electric energy that is expected to be consumed may fluctuate violently because of the difficulty to control the new energy power generation. So we assume the expectation value of each hour subjects to normal distribution. The fluctuation of battery that is exchanged in each hour is relatively smooth, so we suppose it is evenly distributed.

We begin our simulation with the following given value: $N = 25$, $c = 12$, $\alpha = 0.99$, $B = 2$, $v \sim U[1, 9]$, $\sigma_v \sim U[0.3, 0.6]$, $u \sim N(5, 25)$, $\sigma_u \sim U[0.5, 1]$.

We proceed to show our experimental results. We first describe the performance of grid regulation only using EV battery swapping stations, then explain how the combination of battery swapping stations and data centers affect the system performance.

B. Results without DVFS

1) *the effect of B*: Fig.2 shows the value of L of different strategy under the different value of B . The effect of baseline is unchanged since station does not discharge batteries in baseline. The effect of PRES changes little, while two other lines change a lot. However, as $1/B$ changes from $1/2, 1/4, 1/8$ to 0, the EPM of EXPS and bestline adds outstandingly, while other strategies remain relatively stable. In conclusion, as the existence of discharging ability have a considerable influence on the regulation capacity when we apply the EXPS, it is worthwhile to enable the discharging ability in practice.

2) *the influence of σ_u* : Fig.3 shows the value of L of different lines under the different value of σ_u . We can know that the value of L of all lines increase slightly, for the difficulty for grid regulation increase as grid become instable itself. And the rank of regulation capacity of each line doesn't change. In other word, EXPS is still the best strategy which can be

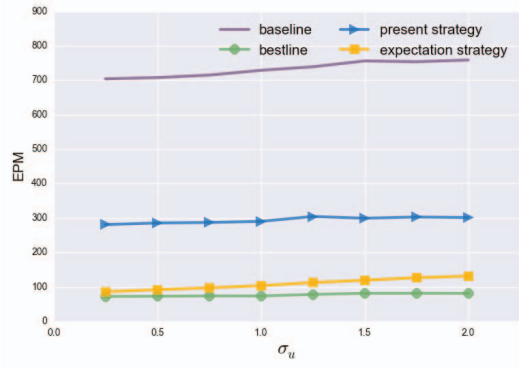


Fig. 3. EPM of varying σ_u .

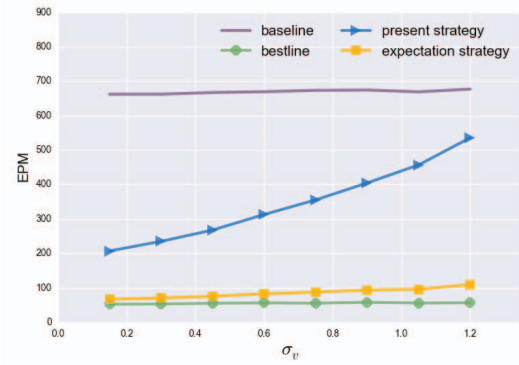


Fig. 4. EPM of varying σ_v .

applied in practice. (Bestline is the optimal solution in theory). However, it is worthwhile to note that the ratio between EXPS and bestline increase 5% as the parameter for sigma added by 0.25. In other word, the effect of EXPS become worse as σ_u increases relative to bestline.

3) *the influence of σ_v* : Fig.4 shows the value of L of different lines under the different value of σ_v . We can know that the value of L of baseline and bestline change slightly, for these lines didn't consider the deviation of battery requirement in particular. And as the value of σ_v increase, the value of L of PRES and EXPS increase for the restriction on grid regulation strengthens. What's more, the effect is more obvious in PRES. So EXPS is still the best strategy which can be applied in practice.

4) *present strategy vs mild strategy*: Fig.5 shows that when β equals to 1, mild strategy (MILS) equals to PRES. We can conclude that PRES is better than mild strategy from the above figure.

5) *expectation strategy vs deviation strategy*: Fig.6 show than when d equals to 0, deviation strategy (DEV) equals to EXPS. The above figure shows the value of L of MILS under the different value of β . We can know that the best effect of DEV emerges when d equals to 0.25. And the ratio of L between EXPS and DEV is 1.004 : 1. Considering the slight difference of L between them, we do not need replace EXPS

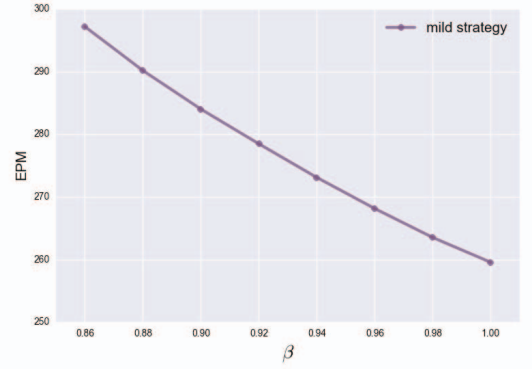


Fig. 5. EPM using mild strategy.

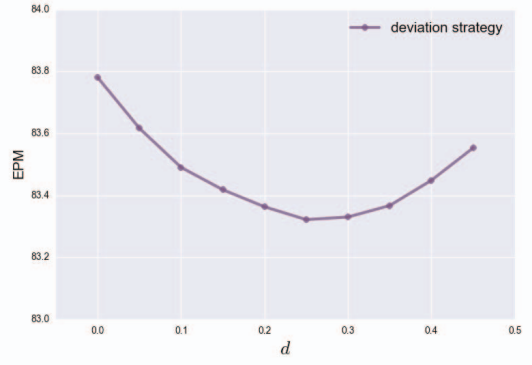


Fig. 6. EPM using deviation strategy.

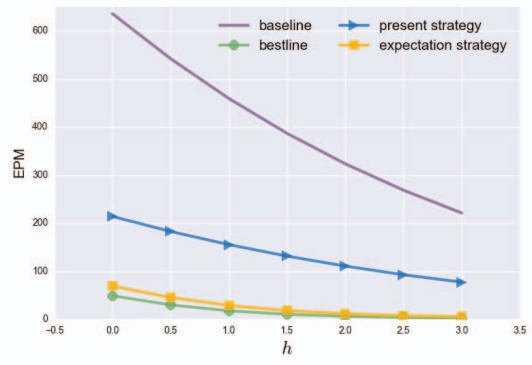


Fig. 7. EPM leveraging data center servers.

by DEV. So there may be another method to get a better effect under DEV. Our group will give a more detailed test to research the strategy when we get actual data.

C. Results with DVFS

In the previous discussions, we treat the regulation capacity of servers in data center as the cost in our objective function. However, the cost may be ignored in some cases. For example, compared to regulation capacity, the cost can be omitted.

1) *data center servers introduced*: We get the following figure through simulation:

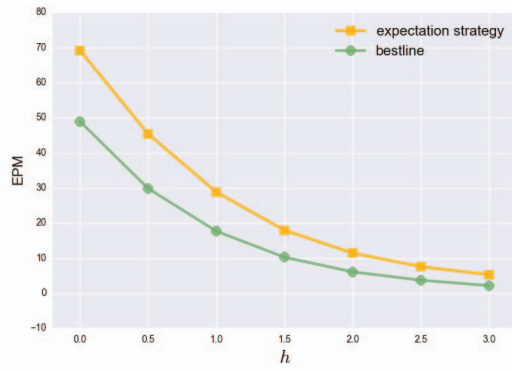


Fig. 8. EPM of varying $h(i)$ using expectation strategy vs. bestline.

Fig.7 shows the value of EPM with data center servers introduced with different value of $h(i)$. We can know that the value of L of each line decline considerably with the regulation capacity of data center improved. And EXPS is still the best strategy which can be applied in practice. And its effect is close to bestline.

2) *expectation strategy with DVFS*: Fig.8 shows the EPM under EXPS and bestline with different value of $h(i)$. We can see that the value of L decline more slowly when the regulation capacity of data center servers grows. Also, the difference between EXPS and bestline decrease as data center servers' regulation capacity improves, and become very close to each other. Overall, the effect of EXPS is close to bestline.

D. Discussion

Simulation results demonstrate that the EXPS always performs better than the PRES, which is the provisional best strategy we can apply. However, the DEV, which is a modified EXPS, has potential to be considerably better than the EXPS if we find a better modified method. Note that the considerations of battery charging rate changes along with its available electric energy and the loss of battery discharging are beyond the scope of this paper. In addition, taking the influence of UPS in data centers into account will be another piece of our future work.

V. CONCLUSION

The combination of data center servers and electric vehicles (EVs) for grid regulation has attracted more attention in recent years. In this paper, we describe a strategy that can be applied to grid regulation involved with battery swapping station coordinating with data center servers. We also prove that the effect of this strategy is close to the best effect in theory no matter how parameters change. Simulation results demonstrate that our solution reduces the imbalance of grid by 80 ~ 85% compared with the baseline approach, and by 50 ~ 70% compared with the other alternative approach.

VI. ACKNOWLEDGEMENT

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